Rukhama Tent House Attendance

System Guide

How the system works, the technology it uses, and what every file does

Live app	https://rukhama-attendance.vercel.app
Source code	https://github.com/durvesh-24/rukhama-attendance
Database	Supabase project *Rukhama Tent Attendance* (Mumbai, ap-south-1)
Hosting	Vercel project *rukhama-attendance*
Document date	16-10-2026

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1. What the system does

Rukhama Tent House runs a per-day labour business. The number of workers changes every month (for example 20 one month and 40 the next), and people join, leave and come back. This web app lets the owner (the **admin**) record all of that and see clear reports. It works on phones and laptops.

- **Employees** - add, edit and photograph workers. Record each **join** and **exit** with a date and a note. Exited workers can **rejoin**; every join/exit period is kept forever.
- **Attendance in shifts** - each day has **Shift 1** (8:00 AM - 1:00 PM) and **Shift 2** (3:00 PM - 8:00 PM), each marked Present / Absent / Leave, plus an optional **Night shift** (9:00 PM onwards) for those who worked it.
- **Dashboard** - today's numbers per shift, a monthly chart and a 12-month headcount trend.
- **Employee report** - pick one worker and a month (or range) to see charts, a calendar, night-shift dates, leave dates and a day-by-day table, and download it as Excel.
- **All reports** - the same for all (or chosen) workers, with an Excel file of four sheets.
- **Admin only** - there is no public sign-up. Only accounts listed as admins can log in.

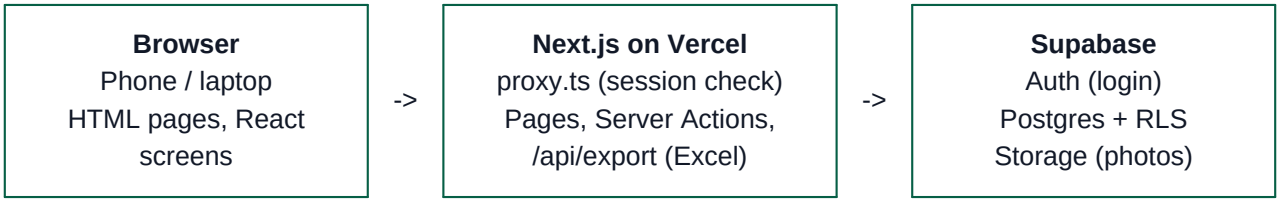
The app does not calculate wages or pay (removed on request). It records attendance only.

2. Technology stack

Layer	Technology	Version	What it is used for
Framework	Next.js (App Router, Turbopack)	16.4	Pages, server rendering, Server Actions (form saves), API routes, the proxy that guards every request.
UI library	React	19.3	Interactive screens (attendance marking, dialogs, filters).
Language	TypeScript	5	Type-checked code; database types are generated from the real schema.
Styling	Tailwind CSS	4	Responsive layout (sidebar on laptop, bottom bar on phone), light/dark mode.
Components	shadcn/ui (Radix UI) + lucide-react icons	radix-nova	Buttons, dialogs, selects, tables, badges, menus.
Charts	Recharts	3.10	Dashboard and employee-report bar / pie / line charts.
Excel	ExcelJS	4.4	Builds the .xlsx downloads on the server (colour-coded, frozen panes).
Validation	Zod	4.6	Checks every form and save on the server before it touches the database.
Notifications	Sonner, next-themes	2.0 / 0.4	Toast messages; light/dark theme switch.
Database	Supabase Postgres (PostgREST API)	PG 17 / PostgREST 14	All data, row-level security, SQL functions for reports.
Login	Supabase Auth (email + password)	-	Admin login, sessions in secure cookies, password change.
File storage	Supabase Storage (private bucket)	-	Employee photos, shown through 1-hour signed links.
Supabase client	@supabase/supabase-js + @supabase/ssr	2.118 / 0.12	Talking to Supabase from the server with the admin's cookie session.
Hosting	Vercel	-	Builds and serves the app; auto-deploys when code is pushed to GitHub.
Source control	Git + GitHub	-	Code history at https://github.com/durvesh-24/rukhama-attendance.
Tooling	Supabase CLI, Vercel CLI, ESLint, tsx	-	Database migrations and types, deployments, code checks, the make-admin script.

3. Architecture and request flow

There are three parts: the **browser** (phone or laptop), the **Next.js app on Vercel**, and **Supabase** (database, login and photo storage). The browser never talks to the database directly; every read and write goes through the Next.js server, which acts as the logged-in admin.



What happens on every request

- **1. Proxy** (*proxy.ts*) runs first. It refreshes the login cookie and sends anyone who is not logged in to */login* (API calls get a 401 error instead).
- **2. Admin check** - the protected layout and every page call *requireAdmin()*; every save (Server Action) and Excel download calls *assertAdmin()*. Both ask the database *is_admin()*.
- **3. Database** - queries run as the logged-in user, so Postgres **row-level security** applies again: rows are only returned or changed if *is_admin()* is true. Even a bug in the app could not expose data to a non-admin.
- **4. Response** - the page is rendered on the server and sent to the browser. After a save, the affected pages are refreshed.

Where things run

Runs in the browser	Runs on the server (Vercel)	Runs in the database
Attendance buttons, dialogs, filters, charts, search, theme toggle (files marked "use client").	Page data loading, admin checks, Server Actions (saves), Excel generation, photo upload.	Row-level security rules, report calculations (attendance_summary etc.), atomic employee join/exit/rejoin, data constraints.

4. How each feature works

4.1 Login (/login)

Email + password via Supabase Auth. After a correct password the server checks *is_admin()*; a non-admin is logged out with a message. There is no sign-up page and sign-ups are disabled in Supabase. Logout is in the sidebar (laptop) or top bar (phone).

4.2 Employees (/employees)

- List as a table on laptop and as cards on phone. Search by name, phone or code; filter by status (Active / Exited) and role.
- **Add employee** - name, phone, role (Labour, Driver, Decorator, Supervisor, Other), address, optional photo, **join date** and **joining note**. A code like **RTH-0001** is created automatically. The employee and their first join period are created together in one database transaction (*create_employee*).
- **Edit** - change details or photo at any time (old photo file is removed).
- **Mark exit** - last working day + reason. Closes the current period and sets status to Exited. Nothing is deleted.
- **Rejoin** - new join date (must be after the last exit) + note. Opens a new period and sets status to Active.

4.3 Employee profile (/employees/[id])

Photo and details, a **join / exit history timeline** (each period editable, overlaps are rejected), and a monthly **calendar**: each day shows two bars (Shift 1, Shift 2) coloured green (Present), red (Absent) or amber (Leave), plus an **N** badge for a night shift. Days outside employment are dashed. Tiles show the month's totals; a button opens the Employee report.

4.4 Attendance (/attendance)

- Pick a date (today by default; past dates editable; future dates blocked). Only employees whose join-exit period covers that date are listed - so the list automatically grows and shrinks month to month.
- Three tabs: **Shift 1**, **Shift 2**, **Night shift**. For today the tab matching the current time opens first (before 2 PM Shift 1, before 8:30 PM Shift 2, otherwise Night).
- Shift tabs: large **P / A / L** buttons per person, **Mark all present**, and **Same as Shift 1** on Shift 2. Night tab: tap **Add night** only for those who worked.
- Optional note per person per shift. One **Save** stores every change across all three shifts in a batch (Server Action *saveAttendance*). Un-marking removes that shift's record. The page warns about unsaved changes.

4.5 Dashboard (/dashboard)

Cards: total active, Shift 1 present, Shift 2 present, night shift today, joined this month, exited this month; a warning when today's marks are incomplete. Charts: present per shift for any month, today's split, and the 12-month headcount trend (with joins and exits).

4.6 Employee report (/employee-report)

Choose an employee (active or exited) and one month or a range of up to 12 months. Shows totals (Shift 1, Shift 2, night shifts, days worked, leave, absent), a shifts-worked chart, a breakdown pie, the calendar (single month), the list of **night-shift dates**, the list of **leave dates** (full or half day) and a day-by-day table. **Download Excel** creates a file for that one employee.

4.7 All reports (/reports)

Single month or month range (max 12), all employees or a chosen set. On-screen summary per employee and the join/exit log; **Download Excel** for everyone in the selection. Tapping a name opens that person's Employee report.

4.8 Excel downloads

File	Sheets
All employees Rukhama_Attendance_<Mon-YYYY>_to_<Mon-YYYY>.xlsx	**Attendance** - one row per employee, three columns per day (S1, S2, N), colour-coded, frozen header and name column, "-" when not employed that day. **Summary** - per-employee totals with a TOTAL row. **Night & Leave Dates** - every night shift and leave with date and notes. **Join-Exit Log** - joins and exits in the period with descriptions.
One employee Rukhama_<Code>_<Name>_<period>.xlsx	**Daily** - one row per date: Shift 1, Shift 2, Night, notes. Plus Summary, Night & Leave Dates and Join-Exit Log for that person.

4.9 Settings (/settings)

Shows the admin account and lets the admin change the password. The **current password is required**, so an unlocked phone with an open session cannot be used to take over the account.

Counting rules

- **Days worked** = dates with at least one shift Present (day or night).
- **Leave days** = dates with any shift on Leave; one shift = half day, both shifts = full day.
- Absent and Leave are also counted in **shifts** (a full-day leave = 2 leave shifts).
- An exit date is the **last working day** (inclusive).

5. Database design

Four tables in the *public* schema of Supabase Postgres. All dates are stored as plain dates; "today" is worked out in India time (Asia/Kolkata).

Table	Columns	Rules
admins	user_id (login account), name, created_at	A login is an admin only if its user_id is here. Rows are added only by the make-admin script.
employees	id, employee_code (RTH-0001...), full_name, phone, address, role, photo_path, status (ACTIVE / EXITED), created_at, updated_at (auto); daily_wage, night_wage (unused)	Code is unique and automatic. Cannot be deleted (history protection).
employment_periods	id, employee_id, join_date, join_description, exit_date, exit_description, created_at	One row per join-exit period. At most one open period per employee. Exit not before join. Cannot be deleted.
attendance	id, employee_id, date, shift (MORNING / EVENING / NIGHT), status (PRESENT / ABSENT / LEAVE), note, marked_at; overtime_hours (unused)	One row per employee + date + shift. A NIGHT row must be PRESENT. Indexed by date.

MORNING = Shift 1 (8 AM - 1 PM), EVENING = Shift 2 (3 PM - 8 PM), NIGHT = 9 PM onwards. A private storage bucket **employee-photos** holds photos (JPG/PNG/WEBP, max 5 MB) at *<employee id>/<time>.jpg*.

Migrations (supabase/migrations)

File	What it does
20261016000001_schema.sql	Creates admins, employees, employment_periods, attendance; employee-code sequence; updated_at trigger; indexes.
20261016000002_rls.sql	is_admin() function; turns on row-level security for every table; admin-only policies; removes all anonymous access.
20261016000003_functions.sql	Report functions and the atomic create / exit / rejoin / edit-period functions.
20261016000004_storage.sql	Private employee-photos bucket and admin-only storage policies.
20261016000005_shifts.sql	Switches attendance to shifts (Shift 1, Shift 2, Night), converts any old rows, updates report functions.
20261016000006_harden.sql	Blocks deleting employees and join/exit history; locks down the trigger helper.

6. Database functions

Function	Purpose
is_admin()	True when the logged-in account is in admins. Used by every security rule.
employees_active_on(date)	Employees whose join-exit period covers the date (used by the attendance page and save check).
attendance_summary(start, end, ids?)	Per-employee totals for any date range: Shift 1 / Shift 2 present, night shifts, absent and leave shifts, leave days, days worked.
daily_attendance_counts(start, end)	Per-day counts per shift for the dashboard chart, plus how many were employed that day.
headcount_trend(months, until)	Headcount, joins and exits per month for the trend chart.
create_employee(...)	Creates the employee and the first join period together (all-or-nothing).
exit_employee(id, date, note)	Closes the open period and marks the employee Exited, with date checks.
rejoin_employee(id, date, note)	Opens a new period (date must be after the last exit) and marks Active.
update_employment_period(...)	Edits a period's dates/notes, rejects overlaps, keeps status in sync.
next_employee_code()	Generates RTH-0001, RTH-0002, ... (never truncates past 9999).

7. Security model

- **No public sign-up**; only admins listed in the admins table can use the app.
- **Three layers of checks**: the proxy (logged in?), the server code (admin?) and the database itself (row-level security on every table).
- Anonymous visitors have **no permissions** on any table or report function (verified: HTTP 401).
- **History cannot be deleted** - employees and join/exit periods have no delete permission at all.
- The **secret service key** is never in the website or on Vercel; it is only used on the developer's computer by make-admin.
- **HTTPS only** (HSTS), the site cannot be embedded in other sites (anti-clickjacking), content-type sniffing off, strict referrer and permissions policies.
- Every input is validated on the server with Zod; photos are type- and size-checked; database constraints back this up.
- Changing the password requires the current password. Login attempts are rate-limited by Supabase.
- Search engines are told not to index the site. App dependencies: 0 known vulnerabilities (npm audit, 16-10-2026).

Full details and the handover checklist are in **SECURITY.md** in the repository.

8. File-by-file reference

Every file in the repository. Files under *components/ui* are standard shadcn/ui building blocks.

Root configuration

File	What it does
package.json / package-lock.json	Dependencies and scripts: dev, build, start, lint, typecheck, db:push, db:types, make-admin. Lock file pins exact versions.
next.config.ts	Next.js settings: security headers, hides the X-Powered-By header, ExcelJS as a server package, 6 MB upload limit for photos, Tailwind loader, and lets a phone on the same Wi-Fi reach the dev server.
proxy.ts	Runs before every request (Next.js 16 name for middleware). Refreshes the login session; sends logged-out visitors to /login.
tsconfig.json	TypeScript settings; the @/ import shortcut points to the project root.
eslint.config.mjs	Code-quality rules (Next.js + TypeScript).
components.json	shadcn/ui settings (style, paths, icon library).
.env.example	Template of the environment variables (no real values). Copy to .env.local.
.gitignore	Keeps secrets (.env.local), build output, node_modules, .vercel and .claude out of Git.
README.md	Setup, Supabase and Vercel deployment instructions, business rules.
SECURITY.md	Security review results and the client handover checklist.
AGENTS.md	Note added by Next.js for AI coding tools (points to the bundled Next.js docs). Not used by the app.

app/ - pages and server code

File	What it does
app/layout.tsx	Root HTML: font, metadata (title, no indexing), theme provider and toast notifications.
app/globals.css	Tailwind and shadcn theme colours for light and dark mode.
app/page.tsx	The home URL (/) - redirects to /dashboard.
app/favicon.ico	Browser tab icon.
app/login/page.tsx	Login page; already-logged-in admins go straight to the dashboard.
app/login/login-form.tsx	The email/password form (browser side).
app/login/actions.ts	Server Actions login (checks password, then admin) and logout.
app/(app)/layout.tsx	Protected area wrapper: checks admin, draws the laptop sidebar, phone top bar and bottom navigation.
app/(app)/loading.tsx	Skeleton placeholders while a page loads.
app/(app)/error.tsx	Friendly error screen with Try again.
app/(app)/dashboard/page.tsx	Dashboard data (cards, charts) for today and the chosen month.
app/(app)/attendance/page.tsx	Loads employees working on the chosen date and their existing shift marks.
app/(app)/attendance/actions.ts	saveAttendance: validates, checks each employee was working that date, saves/removes shift marks in batches.
app/(app)/employees/page.tsx	Loads all employees, their latest join/exit dates and photo links for the list.
app/(app)/employees/actions.ts	createEmployee, updateEmployee (with photo upload), exitEmployee, rejoinEmployee, updatePeriod.
app/(app)/employees/[id]/page.tsx	Employee profile: details, month calendar and totals, join/exit history.

File	What it does
app/(app)/employee-report/page.tsx	Employee report: totals, charts, calendar, night and leave dates, day-by-day table.
app/(app)/reports/page.tsx	All-employee report preview (summary table and join/exit log).
app/(app)/settings/page.tsx	Settings page (account info, password form).
app/(app)/settings/password-form.tsx	Change-password form (browser side).
app/(app)/settings/actions.ts	changePassword: verifies the current password, then sets the new one.
app/api/export/route.ts	GET /api/export - Excel file for all/selected employees (admin only).
app/api/export/employee/route.ts	GET /api/export/employee - Excel file for one employee (admin only).

components/ - screen building blocks

File	What it does
app-nav.tsx	Sidebar and phone bottom navigation with the active page highlighted.
attendance-marker.tsx	The attendance screen: date switcher, shift tabs, P/A/L and Night buttons, notes, bulk actions, Save bar.
dashboard-charts.tsx	Dashboard charts (per-shift bars, today pie, headcount line).
employee-report.tsx	Employee report filters (employee, month/range, download) and its charts.
report-filters.tsx	All-reports filters: month/range, employee picker, preview and download buttons.
month-picker.tsx	Previous/next month switcher used on the dashboard and profile.
page-header.tsx	Page title block and the empty-state message box.
theme-provider.tsx / theme-toggle.tsx	Light/dark mode support and the sun/moon button.
employees/employee-list.tsx	Employee table (laptop) and cards (phone), search, filters, row actions, status badge.
employees/employee-form-dialog.tsx	Add / edit employee dialog with photo picker.
employees/status-dialogs.tsx	Mark exit, Rejoin and Edit join/exit period dialogs.
employees/profile-actions.tsx	Profile buttons (Edit, Exit/Rejoin) and the join/exit history timeline.
employees/attendance-calendar.tsx	Month calendar with Shift 1 / Shift 2 bars and night badges.
employees/employee-avatar.tsx	Round photo with initials fallback.
employees/types.ts	Shared shape of an employee row in lists and dialogs.
ui/*.tsx	shadcn/ui primitives: avatar, badge, button, card, checkbox, dialog, dropdown-menu, input, label, select, separator, skeleton, sonner (toasts), table, textarea.

lib/ - shared logic

File	What it does
auth.ts	requireAdmin() for pages (redirects) and assertAdmin() for actions/APIs (throws).
constants.ts	App name, roles, shifts with times and colours, attendance statuses, photo limits.
dates.ts	India-time 'today' and current shift, DD-MM-YYYY formatting, month/day helpers.
validation.ts	Zod rules for employees, exits, rejoins, periods, attendance batches and passwords.
report-data.ts	Reads report filters from the URL and loads summary, attendance and join/exit data (pages through 1000-row limits).
excel.ts	Builds both Excel workbooks (styles, colours, frozen panes, file names).
shift-days.ts	Groups shift rows into one entry per day; checks if someone was employed on a date.

File	What it does
fetch-all.ts	Reads more than 1000 rows by fetching page after page.
action-result.ts	Standard success/error result for Server Actions and friendly error messages.
format.ts	Initials for avatars.
utils.ts	cn() helper for combining CSS classes.

utils/, types/, scripts/, supabase/, public/

File	What it does
utils/supabase/client.ts	Supabase client for the browser (public key only).
utils/supabase/server.ts	Supabase client for the server using the admin's login cookie (RLS applies).
utils/supabase/middleware.ts	Session refresh + redirect logic used by proxy.ts.
types/database.types.ts	TypeScript types generated from the live database (npm run db:types).
scripts/make-admin.ts	npm run make-admin -- email "Name": makes an existing login an admin (uses the secret key locally).
supabase/config.toml	Supabase CLI project settings (used for linking and migrations).
supabase/.gitignore	Keeps Supabase CLI temporary files out of Git.
supabase/migrations/*.sql	The database, in order (see section 5).
public/	Static files folder (currently empty).

9. Configuration and accounts

Variable	Where	Purpose
NEXT_PUBLIC_SUPABASE_URL	.env.local and Vercel	Supabase project address.
NEXT_PUBLIC_SUPABASE_ANON_KEY	.env.local and Vercel	Public (publishable) key. Safe in the browser - security rules protect the data.
SUPABASE_SERVICE_ROLE_KEY	.env.local only	Secret key. Only for make-admin on the developer's computer. Never on Vercel or GitHub.
SUPABASE_DB_PASSWORD	.env.local only	Database password, used by the Supabase CLI for migrations.

Account	What it holds
Supabase	Database, logins, photos. Settings: sign-ups OFF. Free plan pauses after ~1 week idle and has no automatic backups (Pro recommended for business use).
Vercel	Hosting at https://rukhama-attendance.vercel.app. Hobby plan is for non-commercial use (Pro recommended for business use).
GitHub	Source code at https://github.com/durvesh-24/rukhama-attendance (public; contains no secrets).

10. Operating the system

Daily use (admin)

- Open the live link, log in, go to **Attendance**. Mark Shift 1 in the morning, Shift 2 in the evening, and add night shifts at night. Save each time.
- New worker: **Employees -> Add employee** with join date and note. Worker leaves: **Mark exit**. Comes back: **Rejoin**.
- Month end: **All reports** -> choose month -> **Download Excel** (also a good monthly backup).
- Phone: open the link in Chrome -> menu -> **Add to Home screen**.

Making changes (developer)

```
npm install # once, after cloning
npm run dev # local app at http://localhost:3000
npm run lint && npm run build # checks before publishing
git add -A
git commit -m "commit the changes"
git push # Vercel redeploys automatically
```

Database changes

```
# add a new file in supabase/migrations, then:
npx supabase db push # apply to the live database
npm run db:types # refresh TypeScript types
```

Adding another admin

```
# 1. Supabase -> Authentication -> Users -> Add user (tick Auto Confirm)
npm run make-admin -- person@example.com "Their Name"
```

To remove an admin, delete their row from the *admins* table in the Supabase SQL editor (and optionally the login user).

Backups

On the Free plan there are no automatic database backups. Download the monthly Excel files, or upgrade Supabase to Pro for daily backups.

11. Troubleshooting

Problem	Fix
Login says "not an admin"	Run make-admin for that email (the login exists but is not in the admins table).
"Wrong email or password"	Check the details. A forgotten password can be reset by the developer in Supabase -> Authentication -> Users.
Site is slow to start or shows a database error after a quiet week	The free Supabase project paused. Open the Supabase dashboard and click Restore / Resume.
An employee is missing on the attendance page	Their join-exit period does not cover that date. Check the join date, or Rejoin them.
Cannot mark a future date	By design - attendance is only for today or past dates.
Excel download does nothing	Show the preview first (All reports), and make sure you are still logged in.
Changes not visible on the live site	Check the latest deployment in the Vercel dashboard; push again if it failed.